**What is O/S & functions?**

**Operating System (OS):** A system software that acts as an interface between hardware and software, managing computer resources and enabling user applications to run effectively.

**Function:** A specific task or action that something is designed to perform.

**Job:** Every operation or task executed by the computer.

**Category of OS**

single user single task 0/S --- > MS-DOS

single user multitasking O/S --- > Windows client OS

multi user multi tasking O/S --- > Unix/Linux and Windows Server OS

| --- | --- | --- |
| **NT Lineage** | | |
| 11 | 32/64 | 2021 |
| 10 S | 32/64 | 2017 |
| 10 | 32/64 | 2015 |
| 8.1 | 32/64 | 2013 |
| 8 | 32/64 | 2012 |
| 7 | 32/64 | 2009 |
| Vista | 32/64 | 2006 |
| XP | 32/64 | 2001 |

**95 Lineage:**

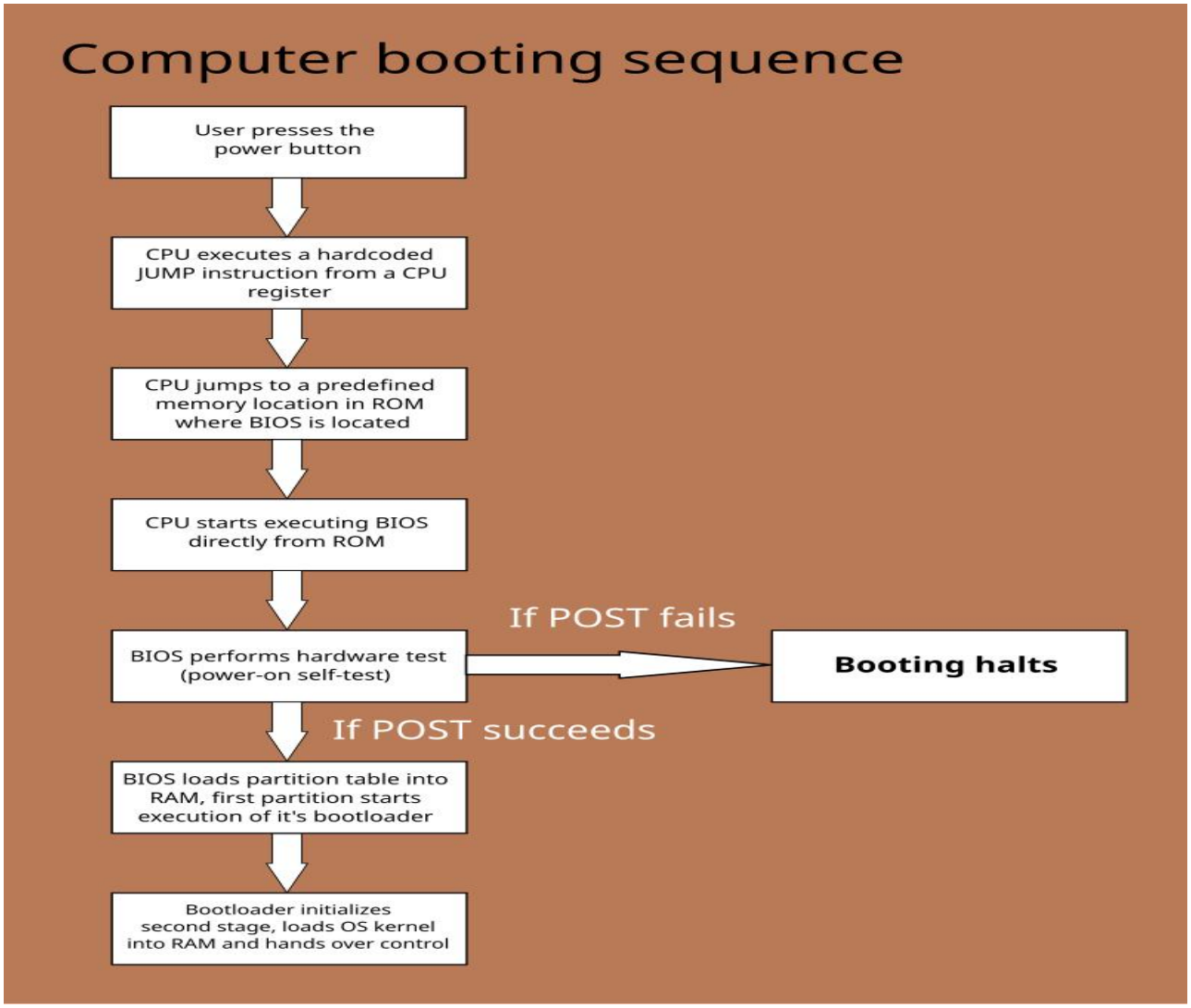
| | | |
|---|---|---|
| ME | 32 | 2000 |
| 98 | 32 | 1998 |
| 95 | 32 | 1995 |

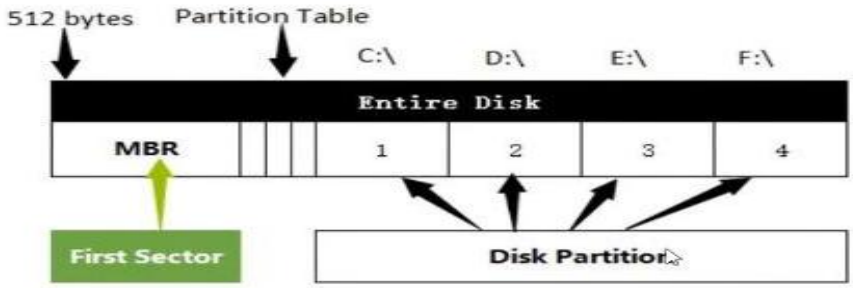
**DOS Lineage:**

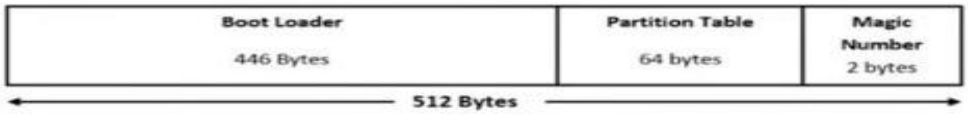
| | | |
|---|---|---|
| 3.11** | 16 | 1992 |
| 3.1 | 16 | 1992 |
| 3.0 | 16 | 1990 |
| 386 | 16 | 1987 |
| 2.0 | 16 | 1987 |
| 1.0 | 16 | 1985 |

**Windows Server Operating systems**

| Windows Server Versions | Word Size (bits) | Year Intro |
|---|---|---|
| **NT Lineage:** | | |
| Server 2022 | 64 | 2021 |
| Server 2019 | 64 | 2018 |
| Server 2016 | 64 | 2016 |
| Server 2012 | 64 | 2012 |
| Server 2008 | 32/64 | 2008 |
| Server 2003 | 32/64 | 2003 |
| Server 2000 | 32/64 | 2000 |






**Magic Byte:** A unique identifier used to verify if the MBR (Master Boot Record) is correct.

**MBR (Master Boot Record):** Contains information about where the operating system is located (boot loader).

**File System:** Organizes and stores data properly on a disk; must be assigned before OS installation.

**OS Edition:** Determines the features and how much hardware (like RAM, CPUs) the OS can support.

**Windows:** A GUI-based OS by Microsoft; not case-sensitive (e.g., "File.txt" = "file.txt").

**Linux:** A case-sensitive OS (e.g., "File.txt" ≠ "file.txt").

**Architecture of Windows:** The layered design of Windows OS that defines how hardware, kernel, user mode, and applications interact.

**Server:** Server is a system that provides services or resources (like files, websites) over a network.

**Client** is a system that requests and uses these services**.**

Together, they form a client-server architecture in a network.

**NIC (Network Interface Card):** A hardware component that connects a computer to a network, allowing it to send and receive data. Acts as an interface between the system and the network.

**MAC (Media Access Control Address):** A unique physical address assigned to a NIC for device identification on a network and operates at the data link layer.

**Network Devices:** Hardware used to connect, route, or secure network traffic (e.g., switch, router, firewall).

- **Switch:** A network device that connects multiple devices to form a local area network (LAN). Routes based on MAC Addresses
- **Router:** Connects different networks; routes data based on IP addresses.
- **Firewall:** Monitors and controls incoming/outgoing network traffic based on security rules.

**Difference between Network & Networking:**

- **Network:** A group of connected devices (computers, servers, etc.) sharing resources.
- **Networking:** The process or practice of connecting devices and managing data transfer between them.

**Types of Networks:**

- **LAN (Local Area Network):** Covers a small area like a home, office.
- **MAN (Metropolitan Area Network):** Covers a city or large campus.
- **WAN (Wide Area Network):** Covers large geographic areas (like the Internet).

**Partition, File System, Permissions, OS**

What is Partition:

Disk partitioning or disk slicing is the creation of one or more regions on secondary storage, so that each region can be managed separately. These regions are called partitions. It is typically the first step of preparing a newly installed disk, before any file system

Windows Partition:

C:\

D:\

E:\

F:\ and etc.

What is file system:

It is method of storing the data in an organized fashion on the disk. Every partition on the disk except MBR (Master Boot Record) and Extended partition should be assigned with some file system in order to make them store the data. File system is applied on the partition by formatting it with a particular type of file system.

Windows File systems:

1. FAT16 FAT32 (File allocation Table)
2. NTFS (New Technology File System)
3. ReFS (Resilient file System)

In windows two types of OS:

1. Client Operating system - Windows95/98/xp/me/vista/7/10/11
2. Server operating system - Windows Server NT/2000/2003/2008/2012/2016/19/2022

Type of OS editions in Server O/S:

1.Standared 2.Enterprise 3.Datacenter

| File System | Max File Size | Max Partition Size | Compatibility | Features | Typical Use Cases |
|---|---|---|---|---|---|
| **FAT16** | 2–4 GB | 2–4 GB | Very wide | Basic, no advanced features | Legacy devices, simple partitions |
| **FAT32** | 4 GB | 8 TB (commonly 2 TB) | Very wide | Simple, no journaling | USB drives, memory cards |
| **NTFS** | 16–256 TB | 256 TB | Windows, limited macOS/Linux | Journaling, permissions, compression | Windows drives, secure data storage |
| **exFAT** | 16 EB | 128 PB | Wide across modern systems | Large file support, cross-platform | External storage, large USB drives |

**File system journaling means** recording changes to files and directories in a dedicated log (called a *journal*) before applying them to the main file system.

**Purpose:**

To prevent data corruption and ensure quick recovery after crashes or power failures

- Install virtual machine setup windows server and 2 windows clients
- Create local users and try to login in both server and client.
- Change computer names.

**Network**

1: TCP/IP and UDP/IP

2: IPv4 - OSI Model

3: Assigning IP Address

Transmission Control Protocol (TCP/IP):

TCP (Transmission Control Protocol) is one of the main protocols of the Internet protocol suite. It lies between the Application and Network Layers which are used in providing reliable delivery services. It is a connection-oriented protocol for communications that helps in the exchange of messages between different devices over a network.

An **IP (Internet Protocol) address** is a unique identifier assigned to each device connected to a network, allowing it to communicate with other devices over the internet or a local network.

User Datagram Protocol (UDP/IP):

User Datagram Protocol (UDP) is a Transport Layer protocol. UDP is a part of the Internet Protocol suite, referred to as the UDP/IP suite. Unlike TCP, it is an unreliable and connectionless protocol. So, there is no need to establish a connection before data transfer. The UDP helps to establish low-latency and loss-tolerating connections establish over the network. The UDP enables process-to-process communication.

OSI Model: It is Conceptual framework to understand how devices talk to each other in a network, divided into **7 layers**.

Application Layer (Layer-7): This is where the user application sits that needs to transfer data between or among hosts. For example – HTTP, file transfer application (FTP) and electronic mail etc.

Presentation Layer (Layer-6): This layer helps to understand data representation in one form on a host to other host in their native representation. Data from the sender is converted to on-the-wire data (general standard format) and at the receiver's end it is converted to the native representation of the receiver.

Session Layer (Layer-5): This layer provides session management capabilities between hosts. For example, if some host needs a password verification for access and if credentials are provided then for that session password verification does not happen again. This layer can assist in synchronization, dialog control and critical operation management (e.g., an online bank transaction).

Transport Layer (Layer-4): This layer provides end to end data delivery among hosts. This layer takes data from the above layer and breaks it into smaller units called Segments and then gives it to the Network layer for transmission.

Network Layer (Layer-3): This layer helps to uniquely identify hosts beyond the subnets and defines the path which the packets will follow or be routed to reach the destination.

Data Link Layer (Layer-2): This layer takes the raw transmission data (signal, pulses etc.) from the Physical Layer and makes Data Frames, and sends that to the upper layer and vice versa. This layer also checks any transmission errors and sorts it out accordingly.

Physical Layer (Layer-1): This layer deals with hardware technology and actual communication mechanism such as signaling, voltage, cable type and length, etc.

• What is NIC ?

A network interface controller (NIC, also known as a network interface card, network adapter, LAN adapter or physical network interface, and by similar terms) is a computer hardware component that connects a computer to a computer

• What is Mac or Physical Address?

The physical address which is also called a media access control or MAC address, identifies a device to other devices on the same local network. it's 48 bit address

Two Versions of Addressing Scheme

– IP version 4 – 32 bit addressing (IPv4)

– IP version 6 – 128 bit addressing (IPv6)

IP Address Classes

• Total IP Addressing Scheme is divided into 5 Classes

– CLASS A - LAN & MAN & WAN

– CLASS B - LAN & WAN

– CLASS C - LAN & WAN

– CLASS D Multicasting

– CLASS E Research & Development

IP Address Classes Range

•CLASS A Range

– 0.0.0.0 - 126.255.255.255

Loop Back Address : 127.0.0.1

• CLASS B Range

– 128.0.0.0 - 191.255.255.255

• CLASS C Range

– 192.0.0.0 - 223.255.255.255

• CLASS D Range

– 224.0.0.0 - 239.255.255.255

• CLASS E Range

– 240.0.0.0 - 255.255.255.255

Octet Format:

• IP address is divided into Network & Host Portion

– CLASS A is written as N.H.H.H

– CLASS B is written as N.N.H.H

– CLASS C is written as N.N.N.H

Private and Public IP Address:

• Private IP Address

– CLASS A 10.0.0.0 - 10.255.255.255

– CLASS B 172.16.0.0 - 172.31.255.255

– CLASS C 192.168.0.0 - 192.168.255.255

• Public IP Address

– Apart from the above specified IP addresses all other IP addresses are Public IP's

Logical Topologies:

1.Ring Topology

2.Mesh Topology

3.Bus Topology

4.Star Topology

**Active Directory:**

AD (Active Directory) simply means a Microsoft service used to manage and control users, computers, and other resources in a network from one central place.

Directory: A database in **Active Directory (AD)** that stores and organizes information about users, computers, groups, and other resources in a network, allowing centralized management and authentication.

**AD DS (Active Directory Domain Services)**: A core service of **Active Directory** that stores directory data and handles authentication, authorization, and management of users, computers, and groups within a **Windows domain network**.

**What AD DS Does**

- **Centralized Authentication and Authorization**: AD DS provides centralized authentication for users and devices within a network, allowing them to log in and access resources based on permissions.
- **Directory of Resources**: AD DS functions as a directory of resources, meaning it organizes and stores information about network objects and makes it easy for administrators to manage user accounts, computers, and other resources.
- **Group Policy Management**: AD DS supports Group Policy, which allows administrators to control settings across all computers in the network from a single point.

2.Purpose of Active Directory?

Provides User Logon and Authentication Services using Kerberos protocol.

• To Centralize and Decentralize the resource management.

• To centrally organize and manage:

– User Accounts, Computers, Groups, Network Resources.

• Enables authorized Users to easily locate Network Resources.

3. What is Domain?

• Domain (in AD): A logical group of computers, users, and resources within a network that are managed centrally under the same Active Directory database and security policies, identified by a unique domain name (e.g., company.local).

• Creating the initial domain controller in a network also creates the domain—you cannot have a domain without at least one domain controller.

• Each domain is identified by a DNS domain name

4. What is a Domain Controller?

• A domain controller is a server that is configured to store a copy of the AD DS directory database (NTDS.DIT) and a copy of the SYSVOL folder.

• All domain controllers except RODCs store a read/write copy of both NTDS.DIT and the SYSVOL folder.

• NTDS.DIT is the database itself, and the SYSVOL folder contains all the template settings for GPOs.

• Domain controllers host several other Active Directory– related services, including the Kerberos authentication service and the Key Distribution Center (KDC).

• Kerberos authentication service is used by User and Computer accounts for logon authentication

• KDC is the service that issues the ticket-granting ticket (TGT) to an account that logs on to the AD DS domain.

5. What is AD DS Logon Process?

5.1. User Account is authenticated to Domain Controller

5.2. Domain Controller returns TGT back to Client

5.3. Client uses TGT to apply for access to Workstation

5.4. Domain Controller grants access to Workstation

5.5. Client uses TGT to apply for access to Server

5.6. Domain Controller returns access to Server

Domain Controller (DC): By default, only the Administrator account can log in; other users are restricted unless configured.

Topics:

1: Security Level Permissions

2: Share Level Permissions

3: Adding Mapped Drives

4: Verifying Access Based Enumeration

What are Permissions?

Permissions define the type of access granted to a user,group, or computer to access resources.

Permissions can be applied to resources such as files,folders, and printers.

-Like: Privilege to read a file, delete a file , or create a new file in folder

Types of Permissions:

Security Level Permissions

Share Level Permissions

1: Security Level Permissions:

• Can be Implemented Only on NTFS partitions.

• Security or NTFS Permissions can be set on Drives,Folders and Files.

• By default, Security permissions will be inherited from its parent drive or folder.

• File permissions override folder permissions.

• Creators of files and folders are their owners.

• Different Security Permissions are

– Full Control, Modify, Read & Execute, Write, List Folder Contents

2: Share Level Permissions:

• It can be implemented on NTFS and FAT partitions.

• It can be set on Drives and Shared Folders but not files.

• What are shared folders?

– Shared folders can be accessed from network.

– When you copy or move a shared folder, the folder will no longer be shared.

– To hide a shared folder, include a $ after the name of the shared folder & users access hidden shared folders by typing the UNC path.

• Different Share Permissions are

– Read, Read/Write.

Effects on NTFS Permissions when Copying or moving files and folders:

• When you move/copy files and folders to a different partition, they inherit the permissions of the destination folder

UNC – universial naming convention

3: Adding Mapped Drives

Mapping a drive simply means to connect a local drive with a specially allocated shared directory or folder on another computer. After a drive has been mapped, you can access the shared resource and treat it as if it's located locally on your own computer.

4: Access Based Enumeration:

• Access Based Enumeration displays only the files and folders that a user has permissions to access.

• If a user does not have read permissions for a folder,windows hides the folder from the users view

5: verify Based Enumeration

Note :- Enumeration permissions has to verify from remote machine using UNC Path

hostname: Command to display the system’s network name.

gpupdate: Updates Group Policy settings on a Windows system.